Quasi-Orthogonality and Polysemy: The Limits of Quantum Analogy in Large Language Models

Karl Svozil[1, *]

[1] *Institute for Theoretical Physics, TU Wien (Vienna University of Technology), Wiedner Hauptstraße 8–10/136, 1040 Vienna, Austria*

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We analyze two geometric aspects of meaning-sensitive behavior in next-token models. First, high-dimensional geometry permits exponentially many nearly orthogonal feature directions at a fixed tolerance, although their residual overlaps limit simultaneous readout. Second, polysemy requires us to distinguish classical convex mixtures of sense vectors, context-dependent occurrence vectors, quantum coherence, and an optional shared-projector incidence construction over sense-labelled bases. For ordinary transformers, the quantum comparisons are structural and deliberately limited.

I. FROM FORMAL PREDICTION TO MEANING-SENSITIVE BEHAVIOR

A large language model (LLM) receives strings and returns strings. Yet it can answer questions, continue an argument, translate a passage, and alter its wording when the surrounding text changes. Its behavior is therefore sensitive to distinctions that readers describe as semantic.

The question recalls John Searle's Chinese Room thought experiment [1]. A person who does not know Chinese follows formal rules for transforming Chinese symbols and thereby produces answers that appear intelligent to an outside reader. Searle asks whether successful symbol manipulation is sufficient for understanding—whether semantics follows from syntax. We do not try to settle that philosophical question, and fluent behavior will not be used as evidence for the emergence of meaning or even consciousness. Our question is narrower and operational: what internal organization allows calculations on tokens to construct and exploit distinctions that human readers recognize as semantic?

An LLM computation begins by encoding text as vectors. A tokenizer maps a string to token identifiers, and an embedding table maps each identifier to a learned lookup vector. After positional information is supplied, transformer layers use the available sequence context—the preceding tokens in a causal LLM—to construct a context-dependent vector for each token occurrence.

This use of vectors creates a limited formal point of contact with quantum mechanics, where a pure state is represented by a ray in a complex Hilbert space. Between measurements, an isolated system evolves unitarily—essentially by one-to-one reversible linear transformations preserving vector lengths. A projective measurement stochastically yields one of its possible outcomes with probabilities given by the Born rule. In complex Hilbert spaces of dimension at least three, Gleason's theorem [2, 3] shows that a normalized, context-independent probability assignment additive over mutually orthogonal projectors (corresponding to mutually exclusive propositions or outcomes) gives that Born rule.

Against this shared inner-product background, two issues provide a manageable way to examine the analogy and its limits.

First, a fixed-width representation space of dimension d can contain at most d mutually orthogonal directions, yet a model may need to distinguish vastly more properties. By relaxing exact orthogonality to permit small pairwise overlaps, a high-dimensional space can instead accommodate many more than d nearly orthogonal directions. We examine the resulting tradeoff: *quasi-orthogonality* expands representational capacity, but the residual overlaps accompanying this expansion can accumulate as readout cross-talk, thereby limiting access to the encoded distinctions.

Second, one word can have several meanings. The word "bank" may refer to a financial institution, the edge of a river, or a seat. We ask how a single token type and its many occurrences can represent such polysemy. This leads to comparisons with two different quantum ideas: coherent superposition and one lexical projector shared by several sense-labelled orthonormal bases.

The order of exposition is as follows: Section II begins with tokenization and explains why vector representations are useful. Section III develops the abundance–cross-talk tradeoff and a limited comparison with quantum decoherence. Section IV distinguishes contextual polysemy from coherent superposition and introduces the shared-projector construction. Section V relates these comparisons to adjacent literatures. Appendix D, *A minimal quantum attention model as a representational foil*, makes explicit the extra structure used in a phase-sensitive quantum construction. It is not a reinterpretation of ordinary transformer attention, an efficient quantum algorithm, or a claim of computational advantage.

* karl.svozil@tuwien.ac.at

II. FROM TEXT TO VECTORS

A. Tokenization: symbols before geometry

The first operation is *tokenization*. A tokenizer divides a character string into units drawn from a finite vocabulary and replaces each unit by an integer identifier. A token may be a complete word, part of a word, punctuation, a space-bearing fragment, or a byte-level unit. Thus a token is not automatically a word, still less a word's meaning. Subword schemes assemble rare and previously unseen words from reusable fragments [4].

The assigned integer is merely an address. Identifier 7193 is not semantically closer to identifier 7194 than to identifier 20. The identifiers are therefore used as addresses in a table of vectors on which graded comparison and learned transformation are possible.

B. A token vector is a starting point, not a complete meaning

A learned lookup table assigns a real vector to every token identifier. Two occurrences of the same identifier begin with the same table row. The model must also know order: "the dog chased the cat" is not equivalent to "the cat chased the dog." Position is therefore supplied in some form, either by adding positional information or by using another positional mechanism.

The resulting initial vector is not a contextually complete meaning. It has been trained across many occurrences and can reflect what those occurrences share, but later layers construct a new hidden vector for each particular occurrence. The occurrence of "bank" near "loan" can consequently acquire a different hidden state from the same token near "river," although both began from the same lookup row. This distinction between a token *type* and a contextual token *occurrence* will be essential for polysemy.

Successive transformer layers let each token position combine information from the positions available to it, thereby producing context-dependent vectors. During training, prediction errors adjust persistent model parameters. During ordinary response generation those parameters remain fixed, and each selected token is appended to the text before the next token is predicted.

C. Why vectors?

Vectors are useful for several reasons. Their coordinates can be adjusted continuously during learning. Matrices can transform them and inner products can compare them. Vector addition permits several contributions to coexist in one state. Most importantly here, lengths, directions, projections, and subspaces provide an economical language for graded features.

Suppose a unit vector u has been fitted as a direction associated with some property at a chosen layer. Relative to a declared baseline h_0, the scalar $u^{\mathsf{T}}(h - h_0)$ is the signed coordinate of a hidden state h along that direction. It is the coordinate, not the axis itself, that says on which side of the contrast the state lies and how strongly. In a simplified gender example, the one-dimensional subspace $\mathrm{span}\{u\}$ supplies a fitted direction for a gender-associated contrast. The subspace is unoriented: u and $-u$ span the same line. One may orient u from a feminine-marked sample mean toward a masculine-marked sample mean. Together with h_0, this defines the affine axis $h_0+\mathrm{span}\{u\}$. Positive and negative projections then describe opposite sides of the same axis; they are not two orthogonal directions. The baseline or a probe intercept defines the zero. Modern contextual representations may use several related directions rather than one universal axis, and the score can change from sentence to sentence [5, 6]. The superscript T means transpose, so $u^{\mathsf{T}}h$ is the ordinary Euclidean inner product.

This directional interpretation is operational, not metaphysical. It depends on the layer, data, centering, inner product, and extraction method. A probe that reads a property shows that the information is linearly available; it does not by itself prove that the model uses that direction causally. Orthogonality nevertheless has one precise consequence. If two property directions are perpendicular, a component along either direction contributes no cross-talk when the other is read by an inner product. The two directions are therefore *geometrically decoupled for this linear readout.* If one uses the phrase "operational semantic independence," it means only this absence of readout cross-talk. It does not mean that the corresponding concepts are logically unrelated, statistically independent in the data, or causally independent inside the model.

The reason to emphasize vectors is therefore not that meanings are known in advance to be points in a Euclidean space. It is that a trained vector system can discover useful graded distinctions, combine them, and compare them by simple operations. High dimension adds a striking opportunity: far more approximately separated directions can coexist than there are exactly independent coordinates. That opportunity motivates our first central issue.

III. QUASI-ORTHOGONAL DIRECTIONS IN A FINITE-DIMENSIONAL SPACE

A. Exact independence and approximate separation

In a d-dimensional space, no more than d nonzero vectors can be mutually orthogonal. This is elementary linear algebra: mutually orthogonal nonzero vectors are linearly independent, and a linearly independent family cannot contain more vectors than the dimension. If semantic distinctions required exactly perpendicular axes, fixed width would impose a severe limit.

Exact orthogonality is stronger than most readouts require. Let u and v be unit vectors. Their inner product is the cosine of the angle between them. We call them quasi-orthogonal at tolerance ε when $|u^\mathsf{T} v| \le \varepsilon$, where the tolerance is chosen before the comparison. They are not independent coordinates; they are approximately separated directions.

High-dimensional geometry makes such directions surprisingly abundant. For a uniformly random unit vector in d real dimensions, the overlap with a fixed unit direction is typically of size $1/\sqrt{d}$. Most of the sphere is therefore concentrated near the equator perpendicular to any chosen direction. Lévy's concentration principle expresses this typicality. The Johnson–Lindenstrauss lemma approaches the same phenomenon from another side: a finite collection of points can be placed in a dimension proportional to the logarithm of the number of points while approximately preserving all pairwise distances. Read in the constructive direction, these results show that at any fixed nonzero angular tolerance a d-dimensional representation space admits at least exponentially many quasi-orthogonal directions [7–9].

The word *exponential* is important, but so are its qualifications. It counts directions available at the chosen tolerance, not a second vector-space dimension. It is a geometric possibility, not proof that a trained model uses all those directions. The allowed overlap must be stated, and an exponentially large dictionary is not automatically readable when many of its elements are active at once. Appendix A introduces the notation, and Appendix B gives the concentration, union-bound, and Johnson–Lindenstrauss calculations.

B. The price of abundance: readout cross-talk

The simplest calculation already shows the access problem. Suppose a hidden state contains two unit feature directions,

$$h = au + bv. \tag{1}$$

Reading the first feature by an inner product gives $u^\mathsf{T} h = a + b(u^\mathsf{T} v)$. The desired coefficient is a; the second term is cross-talk from the other feature. Exact orthogonality removes it, while quasi-orthogonality bounds its size by $\varepsilon|b|$.

Many small unwanted terms can nevertheless accumulate. If k comparably sized, random-like features are active and their signs do not align systematically, their mean squared errors add as in the Pythagorean theorem. The resulting root-mean-square (RMS) cross-talk is of order $\sqrt{k/d}$ relative to one unit-scale target. This is an average random-code benchmark, not a guarantee for every learned dictionary. In the worst case, aligned cross-terms can add rather than cancel.

The useful operating region is consequently task dependent. There should be enough distinguishable directions for the features a task needs, but not so many simultaneously active, strongly overlapping directions that the chosen readout fails. There is no universal numerical boundary determined by d alone. It also depends on which features coactivate, their strengths, the decoder, and the tolerated error. A learned model may use its angular budget selectively, placing greater separation between features that frequently need to be read together.

C. Readout cross-talk and a limited quantum comparison

Small overlaps also occur in quantum theory. When alternatives of a quantum system become correlated with nearly orthogonal environmental states, the off-diagonal terms of the reduced state are multiplied by the corresponding environmental overlaps. High dimension can make those factors very small when the conditional states or their relative dynamics satisfy an explicit typicality assumption; dimension alone is not enough, since the conditional states could remain identical. The author's companion analysis studies this purely geometric contribution to decoherence [10].

For an LLM, quasi-orthogonal feature directions reduce linear-readout cross-talk. In environmental decoherence, nearly orthogonal conditional environmental states reduce off-diagonal terms of the reduced state. Both statements contain small inner products, but only the second describes physical decoherence. An LLM has no required system–environment split, unitary branch dynamics, or partial trace. The common element is algebraic, not dynamical or physical [11].

The word *orthogonal* also has different operational meanings here. For the LLM readout it means geometric decoupling: absence of linear cross-talk in the chosen inner product. Within one quantum measurement context, orthogonal rays represent mutually exclusive outcomes. The common inner-product geometry does not make these interpretations identical.

This qualified comparison is still useful. In both settings, exact orthogonality is an ideal limit; approximate orthogonality can already make alternatives operationally difficult to confuse; and the number of pairwise small overlaps is not the same as the number of alternatives that can be reliably recovered together.

IV. POLYSEMY: ONE FORM, SEVERAL MEANINGS

A. Terms

A *sense* is one conventional meaning of a word form. For example, "bank" may mean a financial institution, the edge of a river, a bench or long seat, or the lateral tilt of an aircraft. We use *polysemy* in the broad computational sense: one word form has more than one sense.

For simplicity, assume that the tokenizer represents "bank" by one token. Its vocabulary entry is the *token type*; one use in a text is a *token occurrence*. The surrounding text that helps identify its sense is the *linguistic context*. A *semantic feature*, such as animacy, is one reusable property that may help characterize a sense; it is not itself a complete word meaning.

B. Static and contextual vectors

A static embedding such as word2vec assigns the same vector to every occurrence of a token type [12]. Under the generative model of Arora and collaborators, this type vector is approximately a frequency-weighted average of hypothetical sense vectors [13]. Retaining three senses of "bank" for the illustration gives

$$\begin{aligned} v_{\text{bank}} \approx\ & p_{\text{finance}} v_{\text{bank,finance}} \\ & + p_{\text{river}} v_{\text{bank,river}} \\ & + p_{\text{seat}} v_{\text{bank,seat}}, \\ & p_{\text{finance}}, p_{\text{river}}, p_{\text{seat}} \geq 0, \\ & p_{\text{finance}} + p_{\text{river}} + p_{\text{seat}} = 1. \end{aligned} \tag{2}$$

Here $v_{\text{bank},s}$ is the hypothetical vector for sense s, and p_s is its relative corpus frequency. Equation (2) is a classical convex combination: it summarizes a token type over the corpus but does not identify the sense of one occurrence. It also does not require the sense vectors to be orthogonal.

A transformer likewise has one initial lookup vector for a token type, but it then computes a hidden vector for each occurrence from the available text. Thus the occurrences of "bank" in "central bank" and "river bank" begin with the same lookup vector but can acquire different hidden vectors. This contextualization need not implement the convex sum in Eq. (2).

C. Why this is not a quantum superposition

In Eq. (2), the coefficients are nonnegative real frequencies that sum to one. In a quantum pure state $|\psi\rangle = \sum_s c_s |s\rangle$, the c_s are complex amplitudes, $\sum_s |c_s|^2 = 1$, and relative phases can change later measurement probabilities. Ordinary embeddings specify neither such phase-sensitive transformations nor a Born-rule measurement. Therefore a static convex mixture is not a coherent quantum superposition, and contextualization is not quantum collapse.

Appendix C displays the pure-state and density-matrix comparison explicitly. Appendix D instead supplies a separate representational foil. Its "bank" calculation shows how a controlled relative phase can affect a Born-rule outcome after coherent preparation and an incompatible readout. The phase is not assigned a linguistic meaning and is not part of ordinary transformer computation.

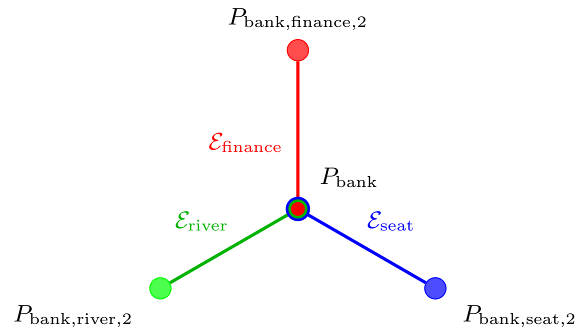

FIG. 1. Shared-incidence schematic for "bank." For each sense s, the basis $\mathcal{E}_s$ contains the lexical projector P_{bank} and the shown projector $P_{\text{bank},s,2}$. Unshown projectors complete each basis. The diagram shows shared incidence, not a vector sum.

D. A shared-projector incidence model

A separate, limited analogy uses incidence rather than addition. Here a *rank-one projective measurement context* means an orthonormal basis, equivalently the associated mutually orthogonal rank-one projectors, which sum to the identity. In dimension $d \geq 3$, different contexts can share one projector while differing in the others.

For the linguistic analogy, let each sense s of "bank" have a context $\mathcal{E}_s$ that contains the same lexical projector P_{bank}. The other projectors specify the basis associated with s. Here s is a linguistic sense and $\mathcal{E}_s$ is the quantum-style context assigned to it, not the surrounding text. This is a representation after s has been assigned, not a rule for inferring s. Unlike Eq. (2), this model places one common element in several bases; it does not average sense vectors.

Orthogonality has different roles in the two settings. For an embedding readout, $u^{\mathsf{T}} v = 0$ removes cross-talk from v when the coefficient along u is read by inner product; it does not imply statistical independence. In quantum theory, mutually orthogonal projectors can represent mutually exclusive outcomes in one measurement context. The analogy therefore uses only the shared incidence structure.

The displayed configuration is Kochen–Specker colorable: assigning value 1 to P_{bank} and 0 to every other projector gives exactly one value 1 in each context. Shared incidence alone therefore does not establish Kochen–Specker contextuality [14, 15].

The shared projector does not select a sense; linguistic context may favor one. For a fixed quantum state, Appendix C shows that the Born probability of the shared projector does not depend on the other projectors that complete its context. Since any unit vector can be completed to many bases, basis completion alone has no empirical content. A direct test would fix a layer, fit r_t and a_t on sense-labelled training occurrences, and then freeze them. On held-out occurrences, the RMS deviation of $r_t^{\mathsf{T}} h_\ell(t \mid x)$ from a_t can be compared with sense-specific coefficients and random directions. Controlling for the initial lookup vector would test whether the anchor is more than residual word-form identity.

V. RELATION TO PRIOR WORK

Three adjacent literatures use related language for different objects. In mechanistic interpretability, *feature superposition* means representing more learned features than there are available dimensions; it does not imply quantum coherence [16]. Contextual-representation and word-sense-disambiguation studies ask whether occurrence vectors distinguish meanings in context [17–20]; we use that distinction but propose no new disambiguation method. Quantum cognition applies Hilbert-space probability to concepts and the mental lexicon [21, 22]; here it motivates a structural comparison, not a claim that ordinary transformers implement quantum dynamics.

VI. CONCLUSION

Searle's Chinese Room asks whether correct manipulation of symbols amounts to understanding. We have not answered that philosophical question. We have instead followed one operational path from tokens to meaning-sensitive behavior.

Token identifiers first become vectors because vectors support graded coordinates, learned transformations, comparison, and combination. Their high-dimensional geometry addresses the first problem considered here. A fixed-width representation space can accommodate exponentially many quasi-orthogonal directions at a chosen tolerance, even though it has only d exactly independent coordinates. This is expressive room, not cost-free access. Residual overlaps create readout cross-talk, especially when many relevant features are active together.

Small overlaps reduce cross-talk in a linear semantic readout. Environmental overlaps can also suppress off-diagonal terms in a quantum reduced state, but only the latter process is decoherence. The two settings share an inner-product pattern, not a physical mechanism.

The discussion of polysemy distinguishes four constructions: a static type vector that may approximate a frequency-weighted convex mixture of sense vectors; context-dependent occurrence vectors; a coherent quantum superposition; and the optional shared-projector incidence construction. The first two are classical representation models, the third is a quantum comparator rather than an ordinary transformer state, and the fourth does not select a sense by itself.

At present, the relation between ordinary transformer LLMs and quantum mechanics is best regarded as a limited analogy of inner-product geometry rather than a common physical or semantic theory. Language-model activations inhabit finite-dimensional real inner-product spaces; standard quantum mechanics represents pure states as rays in a complex Hilbert space and supplements this geometry with observables and the Born rule; composite systems are represented by tensor products. Even shared terms have different meanings: linguistic context is surrounding text, whereas a quantum measurement context as used here is an orthonormal basis, equivalently its rank-one projectors; "interference" here means linear readout cross-talk, whereas quantum interference is phase-sensitive.

Appendix D makes this distinction constructive. Its phase-sensitive result requires coherent complex amplitudes and a Born-rule readout that ordinary transformer attention does not provide. Because the construction takes the scores as available and then compiles the required state preparation, it is a representational foil rather than an end-to-end quantum algorithm; no efficient implementation or speedup is established.

Quasi-orthogonality is nevertheless a genuine common geometric motif. In a language model it offers a possible capacity mechanism for many approximately distinguishable features, subject to cross-talk and empirical confirmation. In quantum theory, near-orthogonality of conditional environmental states can, under appropriate dynamics, suppress reduced-state coherences and thereby contribute to quasi-classical behavior. It does not by itself select a pointer basis, produce definite outcomes, or establish stable classical records. The paper thus identifies geometric resources for semantic discrimination, not a complete theory of machine understanding.

Appendix A: Formal companion to *From text to vectors*

This appendix gives a minimal mathematical version of Sec. II. It distinguishes strings, token identifiers, and vectors before discussing semantic directions.

1. Tokenization and lookup

Let $\mathcal{S}$ be the set of finite character strings. Let $\mathcal{V} = \{1, \ldots, V\}$ be a vocabulary containing V token identifiers. Each identifier is an integer label. A tokenizer is a fixed map

$$\mathsf{Tok} : \mathcal{S} \longrightarrow \bigcup_{T \geq 0} \mathcal{V}^T. \tag{A1}$$

If $\mathsf{Tok}(\sigma) = (t_1, \ldots, t_T)$, then the input string σ has become a sequence of T identifiers. The union in Eq. (A1) allows strings to produce sequences of different lengths. A detokenizer reverses the assembly, subject to the system's stated normalization rules.

Let d_{model} be the architectural width of the model's hidden stream. The notation $\mathbb{R}^m$ means the set of real column vectors with m coordinates. An embedding table is a real matrix $E \in \mathbb{R}^{V\times d_{\text{model}}}$. Its row $E[t_i]$ is the initial lookup vector for identifier t_i. In a simple additive positional scheme,

$$h_i^{(0)} = E[t_i] + p_i, \qquad h_i^{(0)}, p_i \in \mathbb{R}^{d_{\text{model}}}. \tag{A2}$$

Here p_i supplies the position and the superscript (0) means the stage before the first transformer layer. Some architectures incorporate position inside later comparison operations rather than literally adding p_i; the conceptual role is the same. The identifier t_i, the table row $E[t_i]$, and the contextual state produced at a later layer are three different objects.

The geometric analysis need not use all d_{model} architectural coordinates. A declared preprocessing map may center the observed states and remove nearly unused directions. It may also rescale the retained principal directions to a declared common variance, a step called whitening. We use $d \le d_{\text{model}}$ for the rank of the resulting representation space. Any numerical claim involving d must therefore report the model, layer, sample, centering, rank threshold, and inner product used to obtain it.

2. Projection onto a fitted feature direction

Let $u \in \mathbb{R}^d$ be a unit vector, so $u^\mathsf{T} u = 1$. The orthogonal projection of a state h onto the line through u is

$$P_u h = (u^\mathsf{T} h)u, \qquad P_u = uu^\mathsf{T}. \tag{A3}$$

The scalar $u^\mathsf{T} h$ is its signed coordinate. The matrix P_u is called a projector because direct multiplication gives $P_u^2 = P_u$. The remainder $h - P_u h$ is perpendicular to u, since $u^\mathsf{T}(h - P_u h) = 0$. This is the elementary orthogonal projection theorem in finite-dimensional form.

An oriented contrast needs a baseline. Suppose two training samples have means μ_A and μ_B in the same representation space and $\mu_A \neq \mu_B$. Define

$$\begin{aligned} g &= \frac{\mu_A - \mu_B}{\|\mu_A - \mu_B\|}, \quad h_0 = \frac{\mu_A + \mu_B}{2}, \\ \alpha_g(h) &= g^\mathsf{T}(h - h_0). \end{aligned} \tag{A4}$$

The unit vector g orients the contrast, h_0 declares its midpoint, and $\alpha_g(h)$ is the signed coordinate. Substitution shows

$$\alpha_g(\mu_A) = \frac{\|\mu_A - \mu_B\|}{2}, \qquad \alpha_g(\mu_B) = -\frac{\|\mu_A - \mu_B\|}{2}. \tag{A5}$$

Swapping the two groups reverses g and every sign but does not change the unoriented line. A score near zero means that the coordinate along g is near the midpoint; it does not imply that the full vector is near h_0, because a large perpendicular component may remain.

The same construction can illustrate a fitted gender-associated contrast, but it does not define gender and need not capture all gender-related information. A linear classifier provides a related alternative: if its decision boundary is $w^\mathsf{T} h + \beta_0 = 0$ with $w \neq 0$, then $(w^\mathsf{T} h + \beta_0)/\|w\|$ is signed Euclidean distance from that hyperplane. Either construction must be fitted on training examples and evaluated on held-out examples.

Appendix B: Formal companion to the geometric argument

The main text used three counts. The integer d is the exact dimension of the selected vector space. The integer M is the number of feature or code directions in a declared dictionary. The integer k is the number of those directions appreciably active in one state. Only d is a vector-space dimension.

1. Exact orthogonality and quasi-orthogonality

Let $w_1, \ldots, w_M \in \mathbb{R}^d$ be unit vectors. The family is ε-quasi-orthogonal when

$$|w_i^\mathsf{T} w_j| \le \varepsilon \quad \text{whenever } i \neq j, \qquad 0 < \varepsilon < 1. \tag{B1}$$

The absolute value treats a direction and its negative as the same unoriented axis for purposes of overlap.

When $\varepsilon = 0$, the vectors are mutually orthogonal. To see why there can then be at most d, suppose $a_1 w_1 + \cdots + a_M w_M = 0$. Taking the inner product with w_j leaves $a_j = 0$, because every cross-term vanishes and $w_j^\mathsf{T} w_j = 1$. All coefficients are zero, so the vectors are linearly independent. The definition of dimension now gives $M \le d$.

For $\varepsilon > 0$, the vectors need not be linearly independent. The largest possible M becomes a finite spherical-code problem at the chosen tolerance. Its cardinality is sometimes informally called quasi-dimensionality, but no additional linear dimensions have appeared.

2. Why typical overlaps shrink with dimension

For a random quantity Y, $\mathbb{E}[Y]$ denotes its average. For an event A, $\mathbb{P}(A)$ denotes its probability. A unit vector chosen uniformly from the sphere has no preferred direction: rotating the whole experiment does not change its probability law.

Fix a unit vector u and choose X uniformly from the unit sphere in $\mathbb{R}^d$. Rotational symmetry lets us take u to be the first coordinate direction. The overlap $u^\mathsf{T} X$ is then X_1. Symmetry makes all coordinate squares have the same expectation. Since $X_1^2 + \cdots + X_d^2 = 1$, taking expectations gives

$$1 = d\,\mathbb{E}[X_1^2], \qquad \mathbb{E}[(u^\mathsf{T} X)^2] = \frac{1}{d}. \tag{B2}$$

The RMS overlap is the square root of the mean squared overlap, hence $1/\sqrt{d}$. This elementary calculation gives a scale, not a tail probability and not a guarantee for learned vectors.

Lévy's concentration lemma supplies the tail. One convenient consequence is that universal positive constants c and C exist such that

$$\mathbb{P}\{|u^\mathsf{T} X| > \varepsilon\} \le C \exp(-cd\varepsilon^2) \tag{B3}$$

for a fixed unit u and a relevant fixed tolerance. The function $x \mapsto u^\mathsf{T} x$ is 1-Lipschitz because the Cauchy–Schwarz inequality gives $|u^\mathsf{T}(x-y)| \le \|x-y\|$. Lévy's lemma says that a Lipschitz function on a high-dimensional sphere is unlikely to deviate far from its typical value. Here 1-Lipschitz means that changing the input by a distance r changes the function value by at most r [7, 23].

3. From one pair to an exponential family

Choose M directions independently and uniformly. There are $M(M-1)/2$ unordered pairs. The union bound states that the probability of at least one bad pair is no larger than the sum of the bad-pair probabilities. Combining it with Eq. (B3) gives

$$\mathbb{P}\left\{\max_{i\neq j} |w_i^\mathsf{T} w_j| > \varepsilon\right\} \le \frac{CM(M-1)}{2} \exp(-cd\varepsilon^2). \tag{B4}$$

Choose the greatest integer M not exceeding $\exp(c'\varepsilon^2 d)$, where c' is sufficiently small and d is sufficiently large. Then the right side is below one. Therefore at least one family of that exponential size has every overlap within tolerance. This is an existence lower bound and a random-code benchmark, not a universal maximum and not proof that an LLM learns such a code.

Solving the exponent in Eq. (B4) for the overlap tolerance shows that the union-bound crossover is a constant multiple of $\sqrt{\log M/d}$ when $\log M$ is small compared with d. The union bound supplies a high-probability upper scale, not by itself a matching lower law. This remains a null-model estimate rather than a hard capacity wall.

4. The Johnson–Lindenstrauss route

The Johnson–Lindenstrauss lemma states that a set of $M+1$ points can be mapped into $\mathbb{R}^d$ while preserving all squared distances within relative error δ when

$$d \ge C_{\mathrm{JL}} \delta^{-2} \log(M+1), \qquad 0 < \delta < 1, \tag{B5}$$

where C_{JL} is a numerical constant and log is the natural logarithm [8, 9]. This is a theorem about a prescribed finite set, not random semantic features.

To obtain a quasi-orthogonal code, take the origin and M mutually orthogonal standard coordinate vectors e_i in a sufficiently large starting space. Let Φ be a linear Johnson–Lindenstrauss map and define $y_i = \Phi e_i$. Including the origin controls the image lengths. The polarization identity

$$2y_i^\mathsf{T} y_j = \|y_i\|^2 + \|y_j\|^2 - \|y_i - y_j\|^2 \tag{B6}$$

turns preservation of lengths and distances into preservation of inner products. Before normalization their pairwise absolute inner products are at most 2δ; because every squared length is at least $1-\delta$, the normalized overlaps are at most $2\delta/(1-\delta)$. For a fixed target dimension d, choose an integer M satisfying $M+1 \le \exp(c\delta^2 d)$, with c small enough that Eq. (B5) holds. This constructs an exponential-size family.

Lévy concentration, random spherical coding, and the Johnson–Lindenstrauss lemma are thus related views of one high-dimensional phenomenon. Their exponential factors must not be multiplied as though they were independent mechanisms.

5. A deterministic overlap floor

Random coding gives a typical construction. The Welch bound gives a deterministic limitation. Put the vectors into the columns of a matrix $W \in \mathbb{R}^{d\times M}$ and form the Gram matrix $G = W^\mathsf{T} W$. Its entry $G_{ij} = w_i^\mathsf{T} w_j$ records every pairwise overlap. The diagonal entries are one, so the trace of G is M. The matrix has rank at most d and nonnegative eigenvalues.

Let $\lambda_1, \ldots, \lambda_r$ be its nonzero eigenvalues, where $r \le d$. Cauchy–Schwarz gives

$$M^2 = \left(\sum_{\ell=1}^{r} \lambda_\ell\right)^2 \le r \sum_{\ell=1}^{r} \lambda_\ell^2 \le d \sum_{\ell=1}^{r} \lambda_\ell^2. \tag{B7}$$

The sum of squared matrix entries equals the sum of squared eigenvalues for a real symmetric matrix. Removing the M diagonal ones and averaging the remaining entries yields

$$\max_{i\neq j} |w_i^\mathsf{T} w_j|^2 \ge \frac{M-d}{d(M-1)} \qquad (M > d). \tag{B8}$$

This is the Welch bound [24]. It says that once the dictionary is overcomplete, some nonzero overlap is unavoidable. It does not say which pair overlaps or how damaging that pair is for a particular task.

6. Two-feature and many-feature readout

Let a hidden state be represented by

$$h = \sum_{i \in A} a_i w_i + e. \tag{B9}$$

Here A is the active set, a_i are scalar feature coefficients, and e contains components omitted by this chosen dictionary. Reading feature $j \in A$ by correlation gives

$$w_j^\mathsf{T} h = a_j + \sum_{\substack{i \in A \\ i \neq j}} a_i (w_j^\mathsf{T} w_i) + w_j^\mathsf{T} e. \tag{B10}$$

The first term is desired; the sum contains geometric cross-talk. Equation (1) is the case $A = \{1, 2\}$ and $e = 0$.

If the directions are ε-quasi-orthogonal, the triangle inequality gives the worst-case bound

$$\left| \sum_{\substack{i \in A \\ i \neq j}} a_i (w_j^\mathsf{T} w_i) \right| \leq \varepsilon \sum_{\substack{i \in A \\ i \neq j}} |a_i|. \tag{B11}$$

This bound allows every unwanted term to have the same sign. Under a different, explicitly random-like model, assume instead that the cross-terms are centered, have zero cross-moments, and satisfy $\mathbb{E}[(w_j^\mathsf{T} w_i)^2 \mid w_j] = 1/d$. Treat the active set and coefficients as fixed and condition on w_j. Variances then add:

$$\mathbb{E}\left[\left(\sum_{\substack{i \in A \\ i \neq j}} a_i w_j^\mathsf{T} w_i \right)^2 \middle| A, (a_i), w_j \right] = \frac{1}{d} \sum_{\substack{i \in A \\ i \neq j}} a_i^2. \tag{B12}$$

For $k - 1$ interfering features of comparable unit coefficient, the square root is $\sqrt{(k-1)/d}$. This is the RMS law quoted in the main text. Centering or zero cross-moments is essential; an informal claim of uncorrelated contributions would be insufficient without their means being defined.

The contrast between Eqs. (B11) and (B12) explains why no universal Goldilocks boundary follows from d and M alone. A task must also specify active coefficients, coactivation frequencies, a decoder, and an accepted error. Overlap between two features that never need to be recovered together may have little cost.

7. Linear readout cross-talk and quantum decoherence

We first introduce the notation used in this comparison. A pure quantum state may be represented by a normalized complex column vector; vectors that differ only by an overall phase represent the same physical ray. Dirac notation writes such a vector as a *ket* $|v\rangle$ and its conjugate transpose as the *bra* $\langle v|$. Thus $\langle v|w\rangle$ is an inner product, whereas $|v\rangle\langle w|$ is a matrix called an outer product. A density matrix is a positive semidefinite matrix of trace one; the trace is the sum of its diagonal entries. The partial trace over an environment is the linear operation that discards the environmental coordinates while preserving all expectation values of measurements made only on the system.

For comparison with physical decoherence, let $\{|s_i\rangle\}$ be orthonormal alternatives of a quantum system. Suppose a system–environment interaction produces the pure joint state

$$|\Psi\rangle = \sum_i c_i \, |s_i\rangle |E_i\rangle, \qquad \sum_i |c_i|^2 = 1. \tag{B13}$$

The normalized $|E_i\rangle$ are conditional environmental states. Observing only the system means taking a partial trace over the environment. The reduced system state is

$$\rho_S = \sum_{i,j} c_i c_j^* \langle E_j | E_i \rangle \, |s_i\rangle\langle s_j|. \tag{B14}$$

Thus every off-diagonal term with $i \neq j$ is multiplied by an environmental overlap. When the relevant $|E_i\rangle$ become nearly orthogonal, those coherence terms become small for system-only observations. This is the standard algebraic core of environmental decoherence [11].

Near-orthogonality of the $|E_i\rangle$ requires a stated assumption about the interaction or branch-state ensemble; high dimension alone does not produce it. Moreover, many small off-diagonal terms can accumulate. Therefore, bounding each pair separately does not by itself show that the whole reduced matrix is close to the matrix obtained by deleting all off-diagonal terms.

Compare Eq. (B14) with Eq. (B10). The quantum coherence term is suppressed by $\langle E_j | E_i \rangle$; the classical readout error is suppressed by $w_j^\mathsf{T} w_i$. Both are small-overlap effects, but only the first follows from a joint quantum state and a partial trace. In the LLM setting the effect is ordinary suppression of linear-readout cross-talk, not decoherence.

8. Conditional tensor-product amplification

One further scaling is mathematically correct but not implied for an ordinary transformer. If a separate model genuinely has n tensor factors, each of fixed local dimension $q \geq 2$, then its exact Hilbert-space dimension is

$$d = q^n. \tag{B15}$$

Substitution into the exponential code bound gives

$$M \geq \exp(c\varepsilon^2 q^n) - 1, \tag{B16}$$

which is doubly exponential in the number of factors n at fixed q and ε, for sufficiently large n. This is a code cardinality, not an additional Hilbert-space dimension. A real code is also a valid subset of the corresponding complex representation space. Standard LLM residual width is not known to equal an accessible q^n tensor-product representation space, so Eq. (B16) is not an ordinary LLM capacity claim.

Appendix C: Formal companion to the polysemy comparison

1. Static and contextual vectors

Let t be a token type that represents a complete word, and let $s \in \{1, \ldots, S_t\}$ index that word's senses. Under the generative model of Arora et al., the population type vector $\bar{v}_t$ is approximately

$$\begin{aligned} \bar{v}_t &\approx \sum_{s=1}^{S_t} \pi_t(s) v_{t,s}, \\ \pi_t(s) &= \frac{f_{t,s}}{\sum_r f_{t,r}}, \qquad \sum_s \pi_t(s) = 1. \end{aligned} \tag{C1}$$

Here $v_{t,s}$ is a hypothetical sense vector and $f_{t,s}$ is the number of occurrences with sense s. This relation is model-dependent. Arora et al. prove the two-sense case explicitly; the displayed multisense form uses the same conditioning argument [13].

A transformer makes the different statement that a learned map F_ℓ produces one state for an occurrence in linguistic context x:

$$h_\ell(t \mid x) = F_\ell(E[t], x) \tag{C2}$$

at layer ℓ, where $E[t]$ is the token-type lookup vector. This map need not be a convex mixture of fixed sense vectors.

2. Quantum pure superposition and incoherent mixture

Using the Dirac notation introduced in Appendix B 7, let $\{|s\rangle\}$ now denote orthonormal quantum alternatives. A normalized pure state is

$$|\psi\rangle = \sum_s c_s |s\rangle, \qquad \sum_s |c_s|^2 = 1. \tag{C3}$$

The coefficients c_s are amplitudes. Their relative signs or phases can affect probabilities after a change of measurement basis.

The density matrix of the pure state is

$$|\psi\rangle\langle\psi| = \sum_{s,r} c_s c_r^* |s\rangle\langle r|. \tag{C4}$$

The terms with $s \neq r$ are phase-sensitive coherences. By contrast, an incoherent mixture of the same quantum alternatives is

$$\rho_{\text{mix}} = \sum_s p_s |s\rangle\langle s|, \qquad p_s \geq 0, \qquad \sum_s p_s = 1. \tag{C5}$$

Equation (C5) has no off-diagonal terms in this basis. The lexical average in Eq. (C1) is neither Eq. (C3) nor Eq. (C5): it is an average of real embedding vectors, with no quantum measurement rule.

3. One projector shared by several sense-labelled contexts

Let $d \geq 3$, let $r_t \in \mathbb{R}^d$ be a unit lexical anchor, and put $P_t = r_t r_t^\mathsf{T}$. For each sense s, choose an orthonormal basis

$$\{r_t, u_{t,s,2}, \ldots, u_{t,s,d}\} \tag{C6}$$

and define $P_{t,s,j} = u_{t,s,j} u_{t,s,j}^\mathsf{T}$. The associated quantum-style context is

$$\mathcal{E}_{t,s} = \{P_t, P_{t,s,2}, \ldots, P_{t,s,d}\}, \qquad P_t + \sum_{j=2}^{d} P_{t,s,j} = I_d. \tag{C7}$$

For the quantum-style reading, these real vectors and projectors are identified with their canonical embeddings in $\mathbb{C}^d$. Different sense-labelled contexts can share P_t and differ in their other projectors. Completing r_t to such bases is automatic; it is not an empirical result. Because each basis spans the same space, its choice alone adds no constraint; the fixed projection below is the substantive part. The resulting star of bases is Kochen–Specker colorable: setting $v(P_t) = 1$ and assigning value 0 to every other projector gives exactly one value 1 in each context. Thus this incidence pattern alone is not a proof of quantum contextuality.

An idealized shared-anchor hypothesis for occurrence states can be written

$$h_\ell(t \mid x) = a_t r_t + \sum_{j=2}^{d} z_{t,s(x),j}(x) u_{t,s(x),j}. \tag{C8}$$

Here x is the linguistic context and $s(x)$ is an assigned sense label; the equation does not infer that label. Its exact form requires the same nonzero a_t for every occurrence of type t. This common coefficient is the testable hypothesis; completing r_t to the bases is automatic.

A direct test fixes a layer ℓ, estimates r_t and a_t on sense-labelled training occurrences, and then freezes them. On held-out occurrences, report the RMS error of $r_t^\mathsf{T} h_\ell(t \mid x) - a_t$ and compare it with a model that allows a separate coefficient $a_{t,s}$ for each sense and with matched random directions. The shared-anchor hypothesis requires a nonzero common coefficient with little held-out penalty relative to the sense-specific model. Repeating the analysis after controlling for the initial lookup vector separates a learned anchor from persistent type identity.

For a fixed quantum state ρ, the Born probability assigned to the shared projector is

$$p(P_t \mid \rho) = \operatorname{tr}(\rho P_t). \tag{C9}$$

No term in this expression names the other projectors that complete the measurement context. Hence the shared projector alone does not select a sense.

Appendix D: A minimal quantum attention model as a representational foil

This appendix gives a minimal quantum attention construction as a *representational foil*. Its purpose is to display the extra mathematical structure used here to make relative phase observable: a complex Hilbert space, coherent unitary state preparation, a tensor-product address–value state, and projective Born readout. The partial-trace output supplies the incoherent comparison. The construction is neither a reinterpretation of ordinary transformer attention nor a proposed efficient quantum algorithm. No quantum speedup or empirical linguistic advantage is claimed.

For each target i, the construction uses an address space $\mathcal{H}_A^{(i)} \simeq \mathbb{C}^i$ with orthonormal basis $\{|j\rangle_A : j = 1,\dots,i\}$ and a semantic or value space $\mathcal{H}_S \simeq \mathbb{C}^V$ with a complete orthonormal vocabulary basis $\{|y\rangle_S : y = 1,\dots,V\}$, where V is the vocabulary size. The complete orthonormal vocabulary basis is an explicit assumption of this toy model. Their joint space is $\mathcal{H}_A^{(i)} \otimes \mathcal{H}_S$; tracing out the address factor leaves a state on $\mathcal{H}_S$.

For each i choose a normalized reference $|0\rangle_A \in \mathcal{H}_A^{(i)}$ and choose $|0\rangle_S \in \mathcal{H}_S$. Encode source position j and form query, key, and value states by unitary matrices:

$$\begin{aligned}
&|\psi_j\rangle = U_{\mathrm{enc}}(x_j)|0\rangle_S, \\
&|q_i\rangle = Q_\theta|\psi_i\rangle, \qquad |k_j\rangle = K_\theta|\psi_j\rangle, \qquad |v_j\rangle = V_\theta|\psi_j\rangle.
\end{aligned} \tag{D1}$$

Here $Q_\theta, K_\theta, V_\theta \in U(V)$ are learned. The complex query–key overlap is the score. Independently formed overlaps are not generally one row of a unitary matrix, so they must be normalized explicitly:

$$\begin{aligned}
c_{ij} &= \langle q_i|k_j\rangle, \quad Z_i = \sum_{r\le i} |c_{ir}|^2, \\
\alpha_{ij} &= \frac{c_{ij}}{\sqrt{Z_i}}, \qquad Z_i > 0.
\end{aligned} \tag{D2}$$

Thus $\sum_{j\le i} |\alpha_{ij}|^2 = 1$. The α_{ij} are amplitudes rather than classical weights, and their relative phases are retained.

Let one prefix-dependent unitary $W_i = W_i(x_{\le i};\theta)$ prepare

$$|\Omega_i\rangle = W_i|0\rangle_A|0\rangle_S = \sum_{j\le i} \alpha_{ij}|j\rangle_A|v_j\rangle_S. \tag{D3}$$

The right-hand side is normalized because the address states are orthogonal and every value state is normalized. It can therefore be extended to an orthonormal basis and chosen as one column of a unitary matrix. Operationally, W_i may be compiled as address-state preparation followed by token-controlled value preparation. In this specification, however, the scores c_{ij} and normalized amplitudes α_{ij} must be obtained before W_i is compiled. No coherent score-computation routine or efficient state-preparation circuit is given.

A projective measurement of the address gives the normalized Born attention weights, without a softmax:

$$p_i(j) = \operatorname{tr}[(|j\rangle\langle j|_A \otimes I_S)|\Omega_i\rangle\langle\Omega_i|] = |\alpha_{ij}|^2 = \frac{|c_{ij}|^2}{Z_i}. \tag{D4}$$

If the address is traced out—whether or not it is first measured and forgotten or completely dephased—the retained value state is simply

$$\rho_{S,i}^{\mathrm{mix}} = \operatorname{tr}_A |\Omega_i\rangle\langle\Omega_i| = \sum_{j\le i} p_i(j)|v_j\rangle\langle v_j|. \tag{D5}$$

Equation (D5) defines the incoherent output of the model. It contains no interference between different addresses. In particular, the partial trace does not produce the coherent sum $\sum_j \alpha_{ij}|v_j\rangle$.

From $|\Omega_i\rangle$, that sum is obtained by a postselected complementary-basis projective readout rather than by tracing out A. Put

$$|+\rangle_A = \frac{1}{\sqrt{i}}\sum_{j\le i}|j\rangle_A, \qquad N_i = \left\| \sum_{j\le i} \alpha_{ij}|v_j\rangle \right\|^2. \tag{D6}$$

When $N_i > 0$, conditioned on the $|+\rangle_A$ outcome, the semantic state and the probability of obtaining it are

$$|\phi_i\rangle = \frac{\sum_{j\le i}\alpha_{ij}|v_j\rangle}{\sqrt{N_i}}, \qquad r_i = \Pr(+) = \frac{N_i}{i}. \tag{D7}$$

This postselected projective "quantum eraser" removes which-address information. Measuring which address was selected and coherently erasing that information are alternative readouts of the same prepared state.

The orthonormal vocabulary basis now gives an ordinary projective Born rule:

$$\begin{aligned}
p_i^{\mathrm{coh}}(y \mid +) &= |\langle y|\phi_i\rangle|^2 = \frac{\left|\sum_{j\le i}\alpha_{ij}\langle y|v_j\rangle\right|^2}{N_i}, \\
p_i^{\mathrm{mix}}(y) &= \operatorname{tr}\big(|y\rangle\langle y|\rho_{S,i}^{\mathrm{mix}}\big) = \sum_{j\le i} p_i(j)|\langle y|v_j\rangle|^2.
\end{aligned} \tag{D8}$$

Both distributions sum to one. The coherent probability contains phase-sensitive cross-terms; the mixture does not. The parameters of the unitary matrices may be trained by minimizing, for example, $-\sum_i \log p_i^{\mathrm{coh}}(y_i^{\mathrm{true}} \mid +)$. A physical implementation must also account for the erasure success probability r_i, for instance by training the joint probability $r_i p_i^{\mathrm{coh}}(y_i^{\mathrm{true}} \mid +)$.

For a diagnostic phase intervention, take two orthonormal value states $|\mathrm{F}\rangle$ and $|\mathrm{R}\rangle$, labelled by the financial and river-edge senses of "bank." Let $\alpha_{\mathrm{F}} = 1/\sqrt{2}$ and $\alpha_{\mathrm{R}} = e^{i\varphi}/\sqrt{2}$. Here φ is only a control parameter: the construction gives it no linguistic interpretation and no rule by which text would set or learn it. The orthogonality of the two sense-labelled states is likewise an assumption of this toy example. Successful address erasure prepares

$$|\phi_\varphi\rangle = \frac{|\mathrm{F}\rangle + e^{i\varphi}|\mathrm{R}\rangle}{\sqrt{2}}. \tag{D9}$$

Within this two-dimensional subspace, use the readout basis $|\pm\rangle = (|\mathrm{F}\rangle \pm |\mathrm{R}\rangle)/\sqrt{2}$. The projective Born probabilities are

$$p_+(\varphi) = \cos^2\frac{\varphi}{2}, \qquad p_-(\varphi) = \sin^2\frac{\varphi}{2}. \tag{D10}$$

Changing φ from zero to π exchanges the two readout probabilities, while the direct $|\mathrm{F}\rangle, |\mathrm{R}\rangle$ populations remain $1/2$. By contrast, Eq. (D5) gives the phase-insensitive state $(|\mathrm{F}\rangle\langle\mathrm{F}| + |\mathrm{R}\rangle\langle\mathrm{R}|)/2$ and therefore $p_+ = p_- = 1/2$. Thus relative phase changes a prediction while direct-path populations are held fixed.

These equations can be simulated classically. They specify states and readouts, not an end-to-end quantum attention algorithm. In particular, no coherent score-computation routine or efficient state-preparation procedure is provided, so the appendix establishes no quantum computational advantage. If instantiated on quantum hardware, W_i, address erasure, and projective measurement would be physical operations subject to ordinary measurement and no-cloning constraints. The required states would have to be freshly prepared from classically specified tokens; no unknown quantum state is copied.

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